RADAR User's Guide[1]

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RADAR is a timesharing system written and in use at the MIT Weather Radar Research Project from 1975 until about 1980. It supported simultaneous radar control and data acquisition, processing, recording, and display, running on a highly modified DEC PDP-8/I. See also the *PDP-8/IXS Simulator User's Guide* and my long personal history.

Load and Start

Load `RADARX.LS` from the repository using File→Load Listing in the PDP-8/IXS simulator. Start at 02000. Running from OS/8 will be available eventually.

On TTY1 you will be asked to enter the date and time. Date is MM/DD/YY. The supported date range was originally 1/1/75 to 1/4/86, but has now been updated to 1/1/26 to 1/4/37. Dates beyond this range are accepted but display wrong. Time is HH:MM:SS, 24-hour format (or just HH:MM). Weather Radar never used daylight savings time, but of course you can enter what you like.

You will then be asked to position the TTY paper, originally for making a paginated log printout. Now just type Enter.

TTY1 will now enter a command loop, accepting commands for radar control, processing, and limited display. Other display commands are entered on TTY2.

Input line editing

Since you're entering commands on a Teletype, Backspace does not delete characters. Instead a backslash is printed for each deleted character. ^U deletes the entire input line. ^O aborts output on TTY1 or TTY2 and returns to the command prompt.

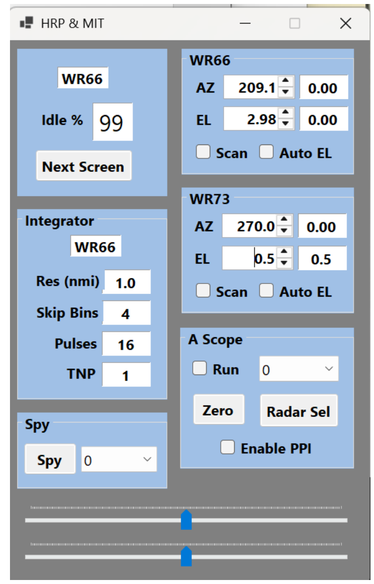
Using the Tektronix 611 Storage Scope

Devices→Tektronix 611 opens a sizeable window that emulates the storage scope that came from HRP and was extensively used at MIT. The scope had the normal storage mode, where plotted points stayed put until the entire screen was erased, and non-store mode, where plotted points disappeared after a short interval and the PDP-8 had to continuously redraw the screen. Mode is determined by the checkbox in the lower right corner of the window.

When TTY1 or TTY2 output is redirected to the scope and the screen fills up, output pauses until Next Screen on the HRP window is clicked.

Using the HRP Window

Devices→HRP opens the window shown below.



Idle% is an emulated 7-segment display that RADAR uses to show the percent of time that the PDP-8 is idle (no process is ready to run, all waiting for I/O).

When characters are being written to the scope and the screen fills up, output pauses until Next Screen is clicked.

The current integrator settings are displayed. Range bins are in nautical miles at the displayed resolution. RADAR converts nmi to km for all commands and displays. Pulses is the number of radar pulses integrated at PRF 250 (pulse repetition frequency).

I don't remember what TNP does.

The Spy controls can examine and modify any memory location while RADAR is running. I used it for diagnostics and, sometimes, to fix problems without having to bring the system down during storms when it needed to stay up. The dropdown selects what the Spy button does. Addresses and data to be written are read from the front panel switches. Memory locations to be examined are displayed in the MQ during idle time—if the CPU is very busy you won't see much. The Load Field command gets the field from SR bits 6-8.

WR66 and WR73 controls allow you or RADAR code to set the position (azimuth and elevation) of the antennas. When Scan is unchecked you can position the antenna with the up/down control, and the PDP-8 can control azimuth scanning speed during recording. When Scan is checked the antenna rotates continuously clockwise and PDP-8 control is disabled. This is very useful with the DI command. Rotation speed is just under one degree per integration period, 4 ms times number of Pulses.

When Auto EL is unchecked you can set antenna elevation with the up/down control. When checked the PDP-8 controls elevation and overrides your setting. WR66 elevation is controlled by a PD control loop running in a dedicated process on the PDP-8. WR73 elevation is set by the PDP-8 using a control loop on the radar itself.

The A Scope controls and the two sliders are described below for the DI command.

Using the PPI Display

A PPI (plan position indicator) is the iconic 2D radar display. The MIT PDP-8 did not have hardware to generate a PPI, but the TI-980 did and RADAR could send live or recorded data to the 980 for PPI display. Devices→PPI opens the display. You can select a monochrome or color display. By 1977 the 980 could generate color displays using a consumer color TV set and a custom interface.

TTY1 Commands

For command names listed here with …, you can enter the given abbreviation or the full name (all characters after the abbreviation are ignored). For example SC… is the abbreviation for the

SCOPE command. For commands without …, the entire command must be entered. Multiple commands can be entered on the command line, separated by semicolons.

Some commands can set or display values. Terminating such a command with ? displays, with = sets. For commands that set multiple values, e.g. RANGE, the values are separated by commas.

Any text enclosed in double quotes is ignored, to provide a comment for the printed log.

One or more commands can be enclosed in () and preceded by a number n. The commands are repeated every n minutes until a terminating condition (e.g. tape full) occurs.

DI

Display integrator. This generates various real-time displays as the antenna is moved, including during recording. DI enters a continuous loop to acquire and process radar data and display them on the scope and optionally the PPI. The scope display is an enhanced A-scope, a graph with the horizontal axis representing distance from the radar and the vertical axis representing amplitude of received power. Here the display is enhanced because the vertical axis has been noise-thresholded, corrected for inverse-square power loss, and converted to dbZ. The DI loop is terminated by typing ^D on TTY1. You can issue SAMPLE and RANGE commands while DI is running.

The scope should be set to non-storage mode for DI, although storage mode can produce interesting displays.

The radar to use for DI is not determined by the WR66 and WR73 commands. Instead you can switch radars with the Radar Sel button.

In addition to the radar graph, DI displays information as chosen by the A Scope dropdown list. If you choose dbZ/KM you get a cross that can be moved with the sliders to display dbZ and km at that point. If you choose Max you can see max dbz values and can reset the max with the Zero button. You can also display time and antenna position.

When Run is unchecked radar data is acquired once and displayed continuously along with updating chosen information. If run is checked new radar data is acquired continuously. You'll notice that the CPU idle time is 0 when run is unchecked—the PDP-8 is continuously updating the display. When checked, and using 32 samples, the 8 is idle about half the time waiting for the integrator.

Currently the emulated integrator gets data from a synthetic image that attempts to emulate rain to the west of the radar. As you move the selected antenna around you'll see the A-scope picking up 1D profiles of the storm.

When Enable PPI is checked the live data is also displayed in the PPI window, generating a 2D picture as the antenna moves.

If you run DI before starting a record command, the A-scope and/or PPI will display what's being recorded.

SC… (SCOPE)

Redirect TTY1 output to the scope. It should be in storage mode. This was very useful given how slow (and noisy) Teletypes were. If TTY2 is using the scope, you'll get a message and output stays on TTY1.

TTY

Redirect TTY1 output to the TTY.

WR66, WR73

Select radar to use for recording. The DI command uses the Radar Sel button instead.

PT66, PT73

Display or set power transmitted in dBM (decibels relative to one milliwatt as measured). Very important for radar calibration, RADAR uses it to adjust dbZ values.

SAM… (SAMPLES)

Display or set the number of pulses to be integrated. Range is 1 – 127, in practice only 16 or 32 were used. The emulated integrator accurately times integration from this value assuming PRF 250. Can be issued while DI is running.

RAN… (RANGE)

Display or set the range covered by the integrator, e.g. RAN=10,150. Values in km. The values will be converted to nmi and the integrator resolution and skip bins will be set to cover at least the given range at the highest possible resolution. Can be issued while DI is running.

TNP

Display/set the mysterious TNP value.

BUGS

Disable clock and timer interrupts for debugging. Not needed on the simulator.

ST… (STATUS)

Display current settings.

AZ… (AZIMUTH)

Display/set the azimuth range for recording, in km, e.g. AZ=220,340.

EL… (ELEVATION)

Attempt to set elevation of selected antenna in tenths degree (no decimal point). The setting only takes effect when Auto El is checked.

ELIST

Set a list of elevations for recording, tenths degree (no decimal point). The default list is 0.5, 1.0, 2.0, 3.0, 4.0, 6.0, 8.0, 10.0, 12.0, 15.0, 20.0, 30.0, 45.0.

LOAD

Set up to use the previously initialized DECtape that is loaded on unit 1.

UPD… (UPDATE)

Update DECtape directory. Note that the tape directory is not automatically updated after a recording, to avoid lots of tape motion time that might delay more recording.

ZERO

Initialize a new DECtape.

REC… (RECORD)

Begin a DECtape recording session. The single numeric argument (default 1) is how many elevations to record for the given azimuth range. When more that one elevation is recorded, the antenna is moved back and forth to minimize recording time. Scan should be unchecked and Auto El should be checked. Recording can be aborted with ^K.

COM… (COMMENT)

Add a comment to the next recording.

TTY2 Commands

SC… (SCOPE)

Redirect TTY2 output to the scope. It should be in storage mode. If TTY1 is using the scope, you'll get a message and output stays on TTY2.

TTY

Redirect TTY2 output to the TTY.

LOAD

Set up to use the previously initialized DECtape that is loaded on unit 2.

DTY

Print the tape directory.

LIST map, azStart-azEnd, elStart-elEnd, binStart (RATE)

List a selection of the rays from the specified map recorded on the tape loaded on unit 2. Map numbers are displayed by the DTY command. Azimuth angles are in degrees, elevation angles in tenths with no decimal point. All of the elements are optional. If map is omitted, the previous one is used. If a start-end selection is omitted, the entire map is used. If (RATE) is given, dbZ values are converted to estimated rainfall rate. BSC and PPI are more useful displays. If the specified map does not exist, RADAR seems to crash, so be careful.

BSC… (BSCAN)

Like LIST, but print a two-color B-scan (if printing is redirected to the scope, only one color is displayed). Same arguments as LIST, except that the bins printed per ray is determined by binStart and the line length on the output device, ASR38 or scope. When using the scope, remember to use the Next Screen button on the HRP window.

PPI

Same arguments as LIST, but make a PPI display. Apparently if binStart is given, the map is read but nothing appears.